

INSA CYBERSECURITY PROGRAM

Networking

Challenge Report & Walkthrough

TryHackMe Networking Concepts

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Target Room: TryHackMe Networking Concepts

Networking

Networking is the practice of connecting computers or building professional relationships.

OSI

The OSI (Open Systems Interconnection) model is a conceptual model developed by the International Organization for Standardization (ISO) that describes how communications should occur in a computer network.

The OSI model is composed of seven layers:

1. physical layer
2. datalink layer
3. network layer
4. transport layer
5. session layer
6. presentation layer
7. application layer

The numbering starts with the physical layer being layer 1, while the top layer, the application layer, is layer 7.

To help remember the layers from bottom to top, we can use a mnemonic such as "Please Do Not Throw Spinach Pizza Away."

layer 1 physical layer

The physical layer, also referred to as layer 1, deals with the physical connection between devices this includes the medium such as a wire, and the definition of the binary digits 0 and 1.

Data transmission can be via an electrical, optical, or wireless signal. Consequently, we need data cables or antennas, depending on our physical medium.

layer2 datalink layer

The data link layer is the second tier of the OSI networking model, responsible for node-to-node data transfer, physical MAC addressing, and framing data across a local physical network.

layer3 network layer

The network layer is layer3 of the OSI model, responsible for logical addressing, routing, and moving data packets across different networks. It takes segments from the transport layer, packages them into packets, and finds the best path to the destination.

layer4 transport

Layer 4, the transport layer, enables end-to-end communication between running applications on different hosts. Your web browser is connected to the TryHackMe web server over the transport layer, which can support various functions like flow control, connection management, and error correction.

Examples: Transmission Control Protocol (TCP) and User Datagram Protocol (UDP).

Layer5 session

The session layer is responsible for establishing, maintaining, and synchronising communication between applications running on different hosts. Establishing a session means initiating communication between applications and negotiating the necessary parameters for the session. Data synchronisation ensures that data is transmitted in the correct order and provides mechanisms for recovery in case of transmission failures.

Layer6 presentation

The presentation layer ensures the data is delivered in a form the application layer can understand. Layer 6 handles data encoding, compression, and encryption.

An example of encoding is character encoding, such as ASCII or Unicode.

Layer7 application

The application layer provides network services directly to end-user applications. Your web browser would use the HTTP protocol to request a file, submit a form, or upload a file.

The application layer is the top layer, and you might have encountered many of its protocols as you use different applications. Examples of Layer 7 protocols are HTTPS, HTTP, FTP, DNS, and more. Don't worry if you are not familiar with all of them.

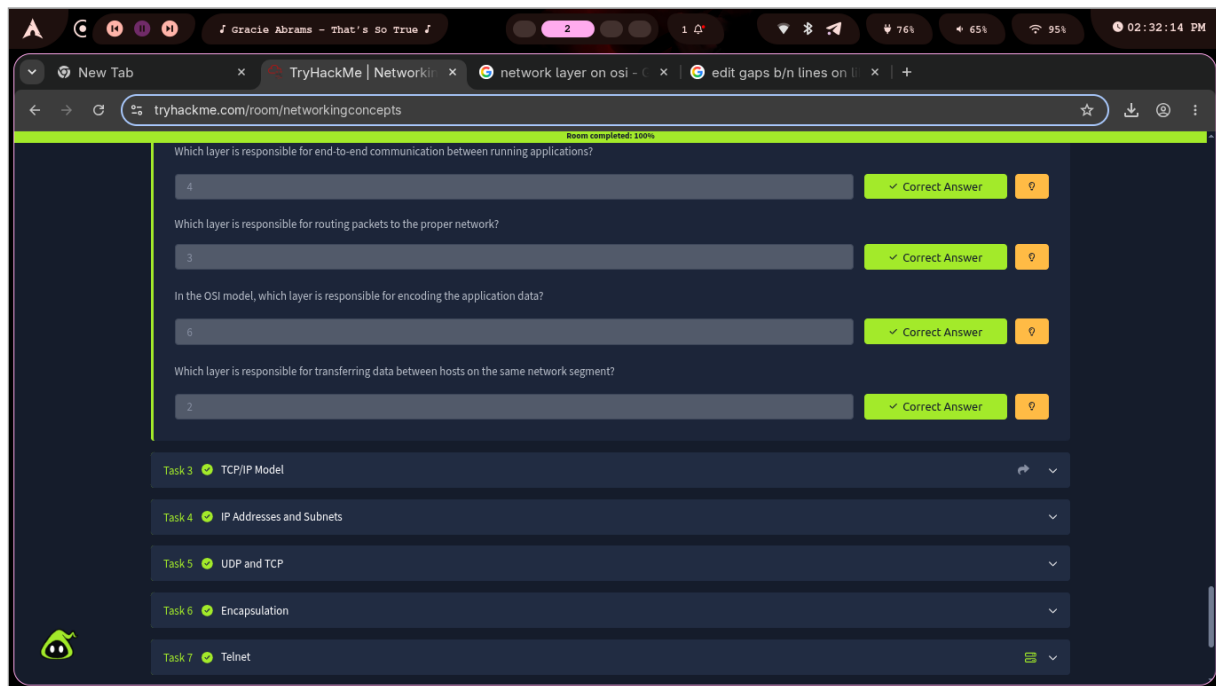


Figure 1: OSI Model — TryHackMe task reference

TCP/IP MODEL

After the OSI model, it is time to study an implemented model, the TCP/IP model. TCP/IP stands for Transmission Control Protocol/Internet Protocol and was developed in the 1970s by the Department of Defense (DoD).

One of the strengths of this model is that it allows a network to continue to function as parts of it are out of service, for instance, due to a military attack. This capability is possible in part due to the design of the routing protocols to adapt as the network topology changes.

From top to bottom, we have:

- **Application Layer:** The OSI model application, presentation and session layers.
- **Transport Layer:** This is layer 4.
- **Internet Layer:** This is layer 3. The OSI model's network layer is called the Internet layer in the TCP/IP model.
- **Link Layer:** This is layer 2.

The screenshot shows a web browser window with the URL `tryhackme.com/room/networkingconcepts`. The page displays a table for the TCP/IP model layers, a list of five layers, and two quiz questions.

Layer	OSI Layer	Protocol
2	Data Link Layer	Link Layer
1	Physical Layer	Ethernet 802.3, WiFi 802.11

Many modern networking textbooks show the TCP/IP model as five layers instead of four. For example, in *Computer Networking: A Top-Down Approach 8th Edition*, Kurose and Ross describe the following five-layer Internet protocol stack by including the physical layer:

- Application
- Transport
- Network
- Link
- Physical

In the following tasks, we will cover the IP protocol from the Internet layer and the UDP and TCP protocols from the transport layer.

Answer the questions below

To which layer does HTTP belong in the TCP/IP model?

Application Layer ✓ Correct Answer

How many layers of the OSI model does the application layer in the TCP/IP model cover?

3 ✓ Correct Answer 0

Task 4 ✓ IP Addresses and Subnets

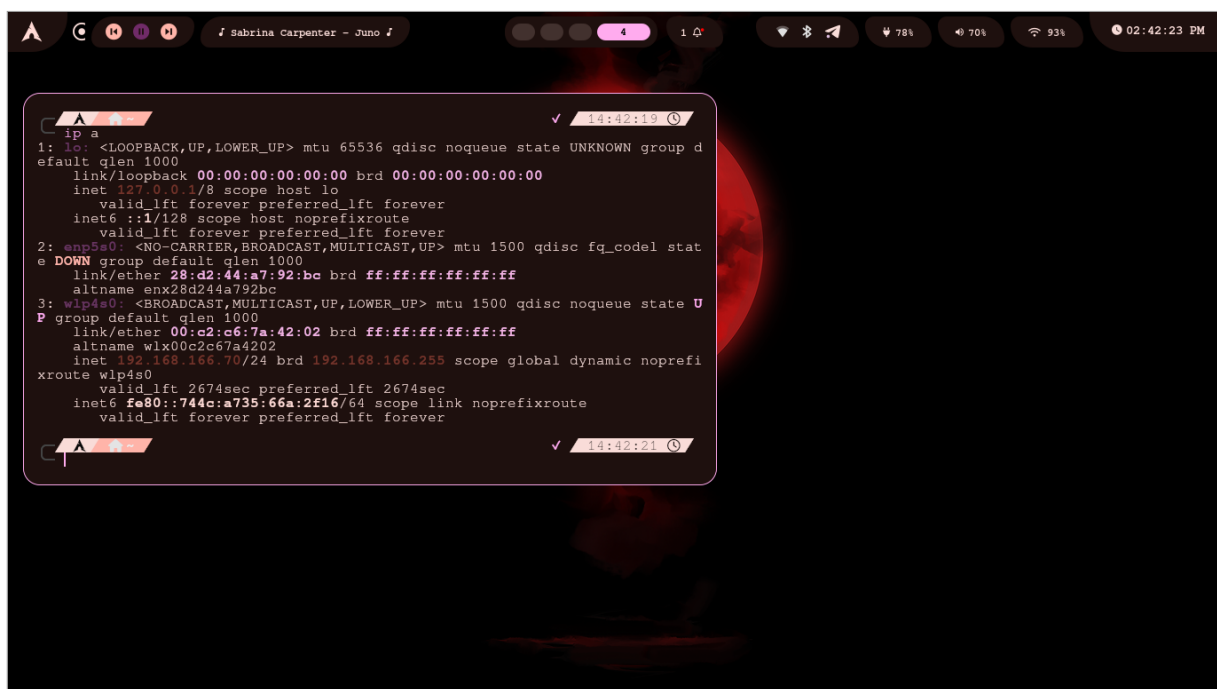
Figure 2: TCP/IP Model — layer breakdown

IP ADDRESS AND SUBNET

An IP address is a unique number assigned to any device connected to a network, acting like a digital home address to help send and receive data.

Looking Up Your Network Configuration

You can look up your IP address on the MS Windows command line using the command `ipconfig`. On Linux and UNIX-based systems, you can issue the command `ifconfig` or `ip address show`, which can be typed as `ip a s`. In the terminal window below, we show `ifconfig`.

A terminal window titled 'Sabrina Carpenter - Juno' showing the output of the 'ip a' command. The output lists three network interfaces: 'lo' (loopback), 'enp5s0' (wired Ethernet), and 'wlp4s0' (wireless LAN). The 'lo' interface has an IP of 127.0.0.1. The 'enp5s0' interface is down and has a MAC address of 28:d2:44:a7:92:bc. The 'wlp4s0' interface is up and has an IP of 192.168.166.70 and a MAC address of 00:c2:c6:7a:42:02. The terminal window has a dark background with a red moon graphic on the right side. The top of the window shows system status icons like battery, Wi-Fi, and Bluetooth, along with the time 02:42:23 PM.

```
ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group d
efault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp5s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel stat
e DOWN group default qlen 1000
    link/ether 28:d2:44:a7:92:bc brd ff:ff:ff:ff:ff:ff
    altname enx28d244a792bc
3: wlp4s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state U
P group default qlen 1000
    link/ether 00:c2:c6:7a:42:02 brd ff:ff:ff:ff:ff:ff
    altname wlx00c2c67a4202
    inet 192.168.166.70/24 brd 192.168.166.255 scope global dynamic noprefi
xroute wlp4s0
        valid_lft 2674sec preferred_lft 2674sec
    inet6 fe80::744c:a735:66a:2f16/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Figure 3: Output of `ip a` on my machine, showing the loopback and wireless interfaces

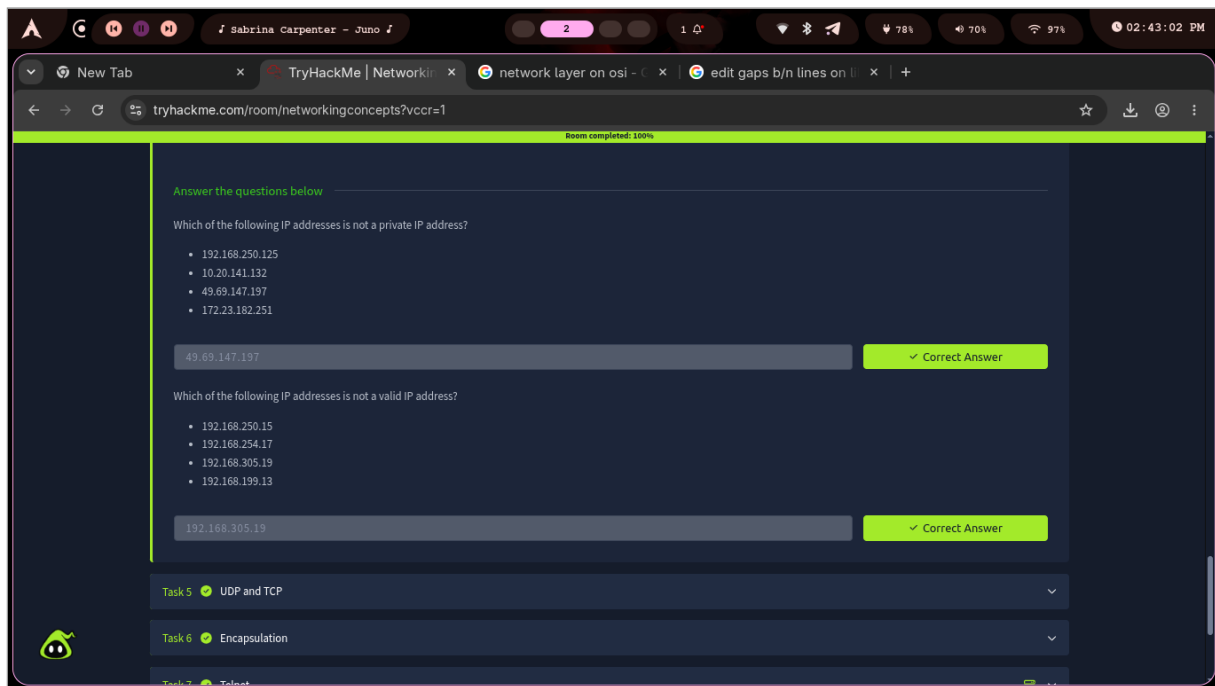


Figure 4: Network configuration — `ifconfig`

TCP and UDP

TCP: The Reliable Handshake

Think of Transmission Control Protocol (TCP) as a registered courier service. It prioritizes accuracy, order, and safety above all else. Before it even thinks about sending your data, it initiates a "3-way handshake" to ensure both the sender and receiver are fully ready to communicate.

Key Attributes

- **Connection-Oriented:** It establishes a dedicated, verified connection before transmitting any data.
- **Guaranteed Delivery:** If a packet of data gets lost in transit, TCP detects the gap and resends it. You will never get a corrupted or incomplete file.
- **Ordered Packets:** Data is broken down into numbered pieces and reassembled at the destination in the exact order it was sent.
- **Smart Traffic Control:** It uses flow and congestion control to dynamically slow down or speed up data delivery based on network conditions, preventing crashes.
- **Heavier Blueprint:** Because it tracks so much data, its header size is relatively large (20 to 60 bytes).

TCP is the go-to protocol when losing even a single piece of data ruins the entire file.

- **Web Browsing (HTTP/HTTPS):** Missing text or code means a broken webpage.
- **File Transfers (FTP/SFTP):** A missing byte in a downloaded app can corrupt the whole program.
- **Email & Messaging (SMTP/IMAP):** Text messages and emails must arrive complete and in order.

UDP: The High-Speed Streamer

User Datagram Protocol (UDP) works entirely differently. Think of it like a live radio broadcast or a postcard. It does not wait for a connection or check if anyone is listening — it just blasts the data out into the network as fast as physically possible. It trades error checking for sheer, unadulterated speed.

Key Attributes

- **Connectionless:** There is no handshake or formal introduction. It starts sending data instantly.

- **Best-Effort Delivery:** It operates on a "fire-and-forget" model. If a data packet drops along the way, UDP ignores it and keeps moving forward.
- **No Packet Ordering:** Data packets take whatever path is fastest and might arrive out of order. The receiving application has to handle the chaos.
- **Maximum Speed:** With no flow control, it pumps data through the pipe without stopping to see if the network is congested.
- **Ultra-Lightweight:** It carries a tiny, fixed header size of just 8 bytes, keeping overhead to an absolute minimum.

Ideal Use Cases

UDP shines in real-time environments where speed is critical, and a tiny bit of data loss goes unnoticed.

- **Online Multiplayer Gaming:** If you experience a split-second drop, you want the game to skip to the present moment, not pause to reload the past.
- **Live Video & Voice Streaming (VoIP):** A dropped packet just means a brief pixelation or a tiny audio glitch; freezing the video to reload a frame would ruin the live experience.
- **Domain Name System (DNS):** Querying an IP address needs to happen instantly, and if it fails, the system simply tries again.

Encapsulation

Encapsulation in the OSI model is the step-by-step process of wrapping data with specific headers and trailers as it moves down from the application layer to the physical layer.

Telnet

Telnet is a text-based network protocol and command-line tool used for bidirectional communication, remote login, and testing network port connectivity. It operates over TCP but it transmits all data and passwords in unencrypted plain text.

How Telnet Works

- **Client-Server Model:** A local client program connects to a remote server or device over a TCP/IP network.
- **Virtual Terminal:** It presents a text command-line interface, letting you interact with the remote host as if you were sitting right in front of it.
- **Port Testing:** Beyond remote logins, administrators use it to test if specific TCP ports (like port 80 for HTTP) are open and responding.

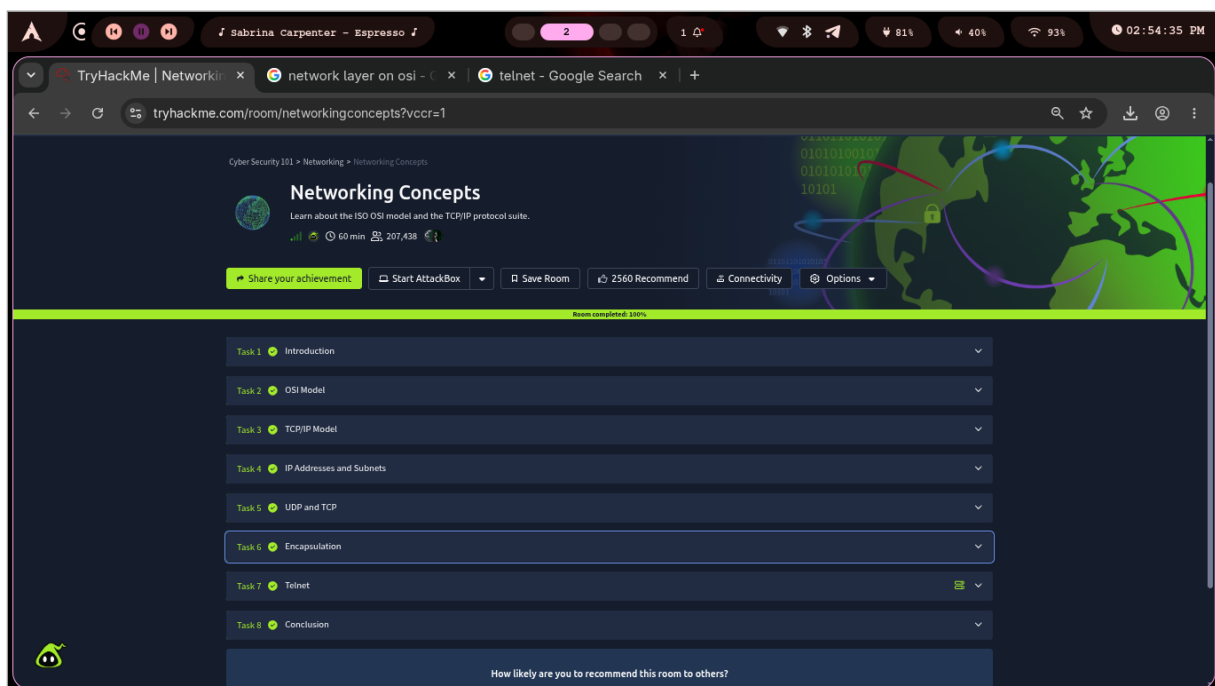


Figure 5: TryHackMe Networking Concepts room — Telnet task

Additional Self-Study Topics

The rooms below (Wireshark, Tcpdump, TLS/SSH/VPN, Nmap) are premium-locked on TryHackMe and I haven't run through them as interactive rooms yet, so there are no screenshots or flags for this section. What follows is what I picked up from watching video walkthroughs on these topics on my own, ahead of getting premium access. I'm including it to show the concepts stuck, not as a claim that I completed these rooms.

Wireshark Basics

Wireshark is a GUI packet analyzer — it captures traffic on a network interface and lets you inspect every layer of a packet, from Ethernet up through the application protocol. Two filter types matter: **capture filters** decide what gets recorded in the first place (applied before capture starts), and **display filters** decide what you see afterward from an already-captured file. A capture filter like `host 10.10.1.1` narrows what's recorded; a display filter like `http` or `tcp.port == 443` narrows what's shown from a full capture. The feature that stood out most from the videos was **Follow > TCP Stream**, which reconstructs an entire conversation between two hosts — that's how you'd recover plaintext credentials sent over something like unencrypted HTTP or FTP.

Tcpdump

Tcpdump is the command-line version of the same idea — useful when there's no GUI available, like over an SSH session on a remote box. It can capture live traffic or read/write `.pcap` files that open later in Wireshark for a deeper look. The commands that came up most: `tcpdump -i eth0` to capture on an interface, `-w file.pcap` to save a capture, `-r file.pcap` to read one back, and filtering by host or port (`tcpdump port 80`). Tcpdump uses the same BPF filter syntax as Wireshark's capture filters, so it's not really a separate skill — more like the same language in a different interface.

Secure Protocols: TLS, SSH & VPN

TLS secures data in transit through a handshake: client and server agree on a cipher suite, the server proves its identity with a CA-signed certificate, and both sides derive a shared session key to encrypt everything after that. It's what turns HTTP into HTTPS. **SSH** gives an encrypted channel for remotely administering a machine — login, commands, file transfer (SCP/SFTP) — with key-based authentication being the stronger option since the private key never has to leave the client. **VPN** builds an encrypted tunnel between a device and a remote network so the device looks like it's local to that network — this is literally how TryHackMe's own VPN connects me to their target machines in the first place.

Core Protocols: TCP/IP Deep Dive

This built directly on the TCP/IP model above, going deeper into how the model actually gets implemented. The main things that clicked from the videos: how a socket (IP address + port) uniquely identifies one specific conversation on a machine, how routing decisions at the Internet layer depend on the destination IP falling inside or outside the local subnet, and how the same handshake-then-transfer pattern in TCP shows up again at the TLS layer — TLS's handshake happens after TCP's, not instead of it.

Nmap Basics

Nmap is the standard tool for host discovery and port scanning — it answers what's alive on a network, what ports are open, and what service/version is behind each one. The core scan types from the videos: `nmap -sn 10.10.1.0/24` for a ping sweep with no port scan, `-ss` for a SYN ("stealth") scan (the default and fastest when run as root), `-st` for a full TCP connect scan when SYN isn't possible, `-sV` for service/version detection, `-O` for OS detection, and `-p-` to scan all 65535 ports instead of just the default top 1000. Nmap results are basically the starting point for everything else in a pentest — you can't target a service you don't know is running.

Challenge Questions

[NOTE: these are my best guess at the room's questions based on the topics covered — double check the exact wording against the actual room before submitting]

1. **Q:** What does OSI stand for? **A:** Open Systems Interconnection.
2. **Q:** How many layers are in the OSI model? **A:** 7.
3. **Q:** What layer of the OSI model do MAC addresses operate on? **A:** Layer 2, the data link layer.
4. **Q:** What layer of the OSI model handles routing between different networks? **A:** Layer 3, the network layer.
5. **Q:** What Layer 4 protocol is connection-oriented and guarantees delivery? **A:** TCP.
6. **Q:** What Layer 4 protocol is connectionless and prioritizes speed over reliability? **A:** UDP.
7. **Q:** What is the process of wrapping data with headers as it moves down the OSI layers called? **A:** Encapsulation.
8. **Q:** What port does Telnet use by default? **A:** Port 23.
9. **Q:** Is Telnet traffic encrypted? **A:** No, Telnet sends everything, including passwords, in plain text.